



Numerical modeling of heat and mass transfer during convective drying of cylindrical quince slices



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ABSTRACT

A numerical model for non-steady heat and mass transfer during convective drying of cylindrical quince slices, with axis parallel to the air flow, is developed. The model is based on the numerical solution of the coupled one-dimensional heat and mass transport equations, assuming moisture transport due to Fick's diffusion, with an effective moisture diffusion coefficient derived by fitting the analytical solution of the Fick's law to experimentally derived drying curves, on the basis of an Arrhenius-type temperature dependence. The necessary convective heat and mass transfer coefficients are obtained from CFD calculations of the turbulent flow field around the slices using a commercial CFD package. A new correlation of the Nusselt number, as a function of Prandtl and Reynolds numbers is proposed for the specific geometric flow configuration. The model is validated against experimental data for different air stream velocities (1 and 2 m/s) and temperatures (40, 50 and 60 °C). The model was found to be robust, computationally efficient and able to capture with sufficient accuracy the time evolution of the temperature and the moisture loss, with a minimum need for experimental adjustment, and hence, is considered suitable from an engineering point of view.

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1. Introduction

Several methods are employed to remove the moisture from organic or non-organic materials, in a variety of industrial applications. In food industry, drying methods include solar, freeze, infrared, microwave and convective drying. Among the various methods, convective drying is one of the most popular, used extensively to dehydrate foods for preservation, transportation and storage purposes (Mujumdar, 2006). In convective drying, the required heat is provided to the food by a hot air stream. Temperature increase causes diffusion coefficient to get higher values resulting to the increase of the drying rate. Drying process is governed by the strong coupling between heat and mass transport, both inside the material and at the interface with the drying air, and is affected by a number of parameters, including the properties of the food

material and the external drying conditions, such as the temperature, the velocity and the relative humidity of the hot air stream.

A considerable number of experiments have been devoted to convective drying of agricultural products. Experiments have been mainly used to assess the drying rates and the total time needed for the removal of the moisture from products, for a range of drying conditions (Velić et al., 2004; Sacilik et al., 2006; Seiedlou et al., 2010; Zlatanović et al., 2013; Babalis et al., 2004; Margaritis et al., 2007; Doymaz, 2009; Erbay et al., 2010). Measurements of the spatial variation of temperature (Wang et al., 1995), as well as of the moisture content (Wang et al., 1995; Aregawi et al., 2013) have been also presented, providing further insight of the dominant physical mechanisms and detailed benchmark basis for the evaluation of the proposed mathematical drying models. Considering, in particular, the spatial distribution of moisture inside the material, the corresponding measurement is difficult to be obtained since it often involves advanced radiography or tomography methods. These methods require specialized equipment and resources, which can be very difficult even to be supplied or accessed, such as for

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Nomenclature

A	total area of product (m^2)	RH	relative humidity of drying air (%)
C	concentration of water vapor (kg/m^3)	t	drying time (s)
c	constant in Nusselt equation	T	absolute temperature (K)
c_p	product specific heat ($\text{J}/\text{kg K}$)	u	velocity (m/s)
d	product diameter (m)	V	product volume (m^3)
D_{eff}	effective moisture diffusivity (m^2/s)	X	moisture content in dry basis (kg water/kg dry product)
D_0	pre-exponential factor of the Arrhenius equation (m^2/s)	z	axial coordinate (m)
D_{va}	diffusivity of water vapor in air (m^2/s)		
DR	drying rate (kg water/kg dry product s)		
e	exponent in Nusselt equation		
E_a	activation energy (kJ/mol)	Greek symbols	
h	heat transfer coefficient ($\text{W}/\text{m}^2 \text{K}$)	a	thermal diffusivity (m^2/s)
h_m	mass transfer coefficient (m/s)	ρ	density (kg/m^3)
$\bar{h}$	weighted average heat transfer coefficient ($\text{W}/\text{m}^2 \text{K}$)	ρ_b	bulk density of product = total mass/total volume (kg/m^3)
$\bar{h}_m$	weighted average mass transfer coefficient (m/s)	ρ_s	density of solid product = dry mass/total volume (kg/m^3)
h_{fg}	water latent heat of evaporation (J/kg)		
k	thermal conductivity ($\text{W}/\text{m K}$)	Subscripts	
L	product thickness (m)	air	drying air
Le	Lewis number	b	back face
M	moisture content in wet basis (kg water/kg product)	d	dry
n	normal vector at the surface	in	initial
Nu	Nusselt number	f	front face
p	pressure (Pa)	fn	final
p_{vs}	saturated water vapor pressure at surface (Pa)	p	product
Pr	Prandtl number	ps	product surface
r	radial coordinate (m)	s	surface
R^2	coefficient of determination	∞	bulk
Re	Reynolds number		
R_g	gas constant, (kJ/mol K)		

example the neutron sources required for neutron radiography (Aregawi et al., 2013).

Mathematical modeling is a very important tool, as it contributes to the better understanding of heat and mass transport, while, when it is simply formulated, it is also a very useful tool in the improvement of the design and the control of the drying process in the food industry. Two main categories of models have been developed, with regard to the major transport mechanisms: in the first, convection by Darcy flow and diffusion is considered, accounting for capillary and pressure driven water flow and diffusion inside the drying material. In the second, Fick's diffusion accounts for the water transport inside the product, by adopting an effective diffusion coefficient that is often obtained experimentally in a semi-empirical manner. An extensive analysis of the different mathematical approaches used for drying modeling can be found in the works of Datta (2007a), Zhang and Datta (2004) and Defraeye (2014).

Diffusion-based models can be further distinguished in two main categories, namely, conjugated and non-conjugated models (Lemus-Mondaca et al., 2011; Defraeye, 2014). In conjugated models, the fully coupled unsteady transport equations for both the drying air and the product are solved (De Bonis et al., 2008; Curcio et al., 2008; Lamnatou et al., 2009, 2010; Marra et al., 2010; Curcio, 2010; Halder et al., 2012; Sabarez, 2012; Kurnia et al., 2013). These models offer high accuracy and extended applicability; however, the complexity of the non-linear mathematical models makes the solution procedure complicated and computationally very demanding. In non-conjugated models, the calculations of the heat and mass transfer inside the product are performed independently from the flow field calculations. The general aspects of the various models proposed for convective drying

of agricultural products using non-conjugated heat and mass transfer are summarized in Table 1. In non-conjugated models, the coupling of the heat equation with the equation of moisture diffusion is often resolved (see the last column of Table 1), by introducing an additional term at the boundary conditions of the heat equation to account for the heat consumed for water evaporation; an approach rather reasonable and preferable from physical considerations point of view (Datta, 2007a, 2007b; Zhang and Datta, 2004). In these models, the heat and mass transfer at the air-product interface is simplified by applying the corresponding convective transfer coefficients (CTC) at the boundary conditions. Several approaches have been proposed in the literature for the derivation of the CTC involving: (i) constant values or correlations for simplified flow configurations such as flow around plates, and cylinders, (ii) time-dependent coefficients based on a zero-dimensional heat and mass balance, (iii) optimization and regression analysis of experimental and numerical data, and (iv) flow field simulations using computational fluid dynamics (CFD).

The above approaches have been adopted in multi or one-dimensional models. In the multi-dimensional models of Hussain and Dincer (2003) and Oztop et al. (2008), constant, empirical values have been used. Heat transfer coefficient obtained by semi-empirical equations for flow over flat plate in combination with mass transfer coefficient calculated on the basis of the heat and mass transfer analogy (Chilton et al., 1934; Incropera et al., 2012) have been adopted in the model of Aversa et al. (2007). Lemus-Mondaca et al. (2013) and Villa-Corralles et al. (2010) incorporated time-averaged heat and mass transfer coefficients calculated by the solution of zero-dimensional heat and mass balance. CTC derived on the basis of CFD simulations have been used in the works of Kaya et al. (2006, 2007b, 2008), Chandra Mohan et al.

Table 1
Summary of studies on convective drying of agricultural products using non-conjugated approach.

Authors	Product	Dimension	HTC ^a	MTC ^b	Evaporation term in heat equation
Ateeque et al. (2014)	Rectangular potato	3D	CFD-Tu ^e	HMTA ^c	Yes
Lemus-Mondaca et al. (2013)	Rectangular papaya	3D	TDC ^d	TDC ^d	Yes
Chandra Mohan and Talukdar (2010)	Rectangular apple	3D	CFD-La ^f	HMTA ^c	No
Esfahani et al. (2014)	Rectangular apple	2D	CFD-La ^f	HMTA ^c	No
Villa-Corales et al. (2010)	Rectangular mango	2D	TDC ^d	TDC ^d	Yes
Kaya et al. (2008)	Rectangular Kiwi	2D	CFD-La ^f	HMTA ^c	No
Oztop and Akpinar (2008)	Rectangular apple & potato	2D	Constant Akpinar and Dincer (2005)	Constant Akpinar et al. (2005)	No
Kaya et al. (2007b)	Cylindrical broccoli & banana	2D	CFD-La ^f	HMTA ^c	No
Aversa et al. (2007)	Rectangular carrot	2D	Empirical Perry et al. (1997) and Bird et al. (1960)	HMTA ^c	Yes
Kaya et al. (2006)	Rectangular apple	2D	CFD-La ^f	HMTA ^c	No
Hussain and Dincer (2003)	Rectangular apple	2D	Constant Dincer (1997)	Constant Dincer (1997)	No
Da Silva et al. (2014)	Cylindrical banana	1D	ORA ^g	ORA ^g	Yes
Barati and Esfahani (2012)	Cylindrical carrot	1D	Empirical Perry et al. (1997)	HMTA ^c	Yes
Da Silva et al. (2012)	Cylindrical banana	1D	–	ORA ^g	No
Barati and Esfahani (2011)	Cylindrical mango	1D	Empirical Perry et al. (1997)	Empirical Janjai et al. (2008)	Yes
Ruiz-López et al. (2004)	Cylindrical carrot	1D	Constant	Constant	Yes
Wang and Brennan (1995)	Rectangular potato	1D	Empirical Earle (1983)	Empirical Burnnett et al. (1962)	Yes
Maroulis et al. (1995)	Potato	1D	ORA ^g	ORA ^g	Yes
Present study	Cylindrical quince	1D	CFD-Tu ^e	HMTA ^c	Yes

^a HTC: heat transfer coefficient.

^b MTC: mass transfer coefficient.

^c HMTA: heat and mass transfer analogy.

^d TDC: time-dependent coefficient.

^e CFD-Tu: CFD-turbulent flow.

^f CFD-La: CFD-laminar flow.

^g ORA: optimization and regression analysis.

(2010) and Esfahani et al. (2014). Considering, in particular the CFD simulations, the air drying stream was considered laminar, except in the recent model of Ateeque et al. (2014) where the SST $k-\omega$ turbulence model (Menter, 1994) was used to analyze the flow field around the product.

Regarding one-dimensional models, CTC related to the thermal and mass Biot numbers (Ratti et al., 1995; Cordova-Quiroz et al., 1996), were used by Ruiz-López et al. (2004). In the work of Wang and Brennan (1995) semi-empirical correlation of the Nusselt number were used. The mass transfer coefficient was taken as a function of the Reynolds number and the water concentration at the surface. Barati et al. (2011, 2012) used also a correlation for the heat transfer coefficient, while the mass transfer coefficient was derived from heat and mass transfer analogy. Heat and mass transfer coefficients derived from optimization and regression analysis using numerical and experimental data have been also presented (Maroulis et al., 1995; Da Silva et al., 2012, 2014).

In this work, a one-dimensional numerical model is developed for the assessment of the non-steady heat and moisture transport within a cylindrical quince slice having its axis parallel to the air flow. Convective drying of quince has been studied in the literature up to some extent (Kaya et al., 2007a; Tzempelikos et al., 2014; Barroca et al., 2012). The coupled heat and mass transport equations are numerically solved using the finite volume method, considering a third kind (Robin) boundary condition to account for the evaporation term. The thermophysical properties of the matter are functions of the moisture content, based on relationships available in the literature, while the effective moisture diffusion coefficient is estimated by fitting the analytical solution of the simplified Fick's diffusion equation (Crank, 1975; Kaya et al., 2007a) to experimental data. A simple mechanistic approach is implemented to model product shrinkage, based on the assumption of volume conservation. A

significant contribution of the present work is the estimation of the heat and mass transfer coefficients for the present flow configuration, which is scarcely considered in the literature. The convective heat transfer coefficient is obtained from steady-state CFD calculations of the turbulent flow field around the slice, incorporating the SST- $k-\omega$ turbulence model. A new correlation for the heat transfer coefficient, as a function of Prandtl and Reynolds numbers is proposed for flow parallel to the axis of a cylindrical slice while the convective mass transfer coefficient is then derived using the heat and mass transfer analogy. New experimental data including the time variation of slice dimensions are presented. Experiments are performed in a laboratory scale convective dryer (Tzempelikos et al., 2012, 2013) for a range of air stream velocities (1 and 2 m/s) and temperatures (40, 50 and 60 °C). Numerical results are presented and compared with measurements and the drying heat and mass transfer mechanisms are discussed in relation to theoretical or experimental findings in the literature.

2. Modeling of heat and mass transfer in quince slices

The mechanisms for moisture removal from the quince are assumed to be (i) the transport of moisture through the product material by liquid diffusion and (ii) the evaporation of water from the surface. No phase change, i.e. evaporation, occurs inside the quince, an assumption suitable for diffusion-based type model, in terms of mass conservation (Datta, 2007a; Zhang and Datta, 2004). The driving force for water diffusion inside the quince is the concentration gradient developed by the evaporation of water on the surface, due to the difference of the water vapor partial pressure between the surface and the air. Liquid water diffusion inside the product is assumed to follow Fick's law. Hot air provides heat energy to the quince, a part of which is consumed for the

evaporation of the water leaving the surface, while the remaining is conducted through the material increasing its temperature. Radiation effects are neglected. Fig. 1 provides a sketch of the computational domain and the parameters involved in the model. Considering the above mechanisms and assumptions, the one dimensional heat and mass transfer equations can be written as follows (Maroulis et al., 1995; Ruiz-López et al., 2004; Wang and Brennan, 1995; Balaban and Piggot, 1988; Tripathi et al., 1973):

$$\frac{\partial \rho_b c_p T}{\partial t} = \frac{\partial}{\partial z} \left(k_p \frac{\partial T}{\partial z} \right) \quad (1)$$

$$\frac{\partial \rho_s X}{\partial t} = \frac{\partial}{\partial z} \left(\rho_s D_{eff} \frac{\partial X}{\partial z} \right) \quad (2)$$

where $z \in [0, L]$ is the axial coordinate (Fig. 1) and t is the drying time.

The following initial and boundary conditions are applied:

$$T(z, t = 0) = T_{in} \text{ and } X(z, t = 0) = X_{in} \quad (3)$$

$$\text{front face : } -k_p \frac{\partial T}{\partial z} \Big|_{z=0} = \tilde{h}_f (T_0 - T_{air}) - h_{fg} \rho_s D_{eff} \frac{\partial T}{\partial z} \Big|_{z=0} \quad (4)$$

$$\text{back face : } -k_p \frac{\partial T}{\partial z} \Big|_{z=L} = \tilde{h}_b (T_{air} - T_L) - h_{fg} \rho_s D_{eff} \frac{\partial T}{\partial z} \Big|_{z=L} \quad (5)$$

$$\text{front face : } -\rho_s D_{eff} \frac{\partial X}{\partial z} \Big|_{z=0} = \tilde{h}_{m,f} (C_0 - C_{air}) \quad (6)$$

$$\text{back face : } -\rho_s D_{eff} \frac{\partial X}{\partial z} \Big|_{z=L} = \tilde{h}_{m,b} (C_{air} - C_L) \quad (7)$$

The vapor concentration at the surfaces is calculated from the ideal gas law (Eqs. (8) and (9)), as a function of the water activity a_w , using the formula proposed by Chen (1971) (Eq. (10)), where k_1 , k_2 and k_3 are constants obtained experimentally by Noshad et al. (2012):

$$C_0 = 2.1667 \times 10^{-3} \times a_w \times \frac{p_{vs}(T_0)}{T_0} \quad (8)$$

$$C_L = 2.1667 \times 10^{-3} \times a_w \times \frac{p_{vs}(T_L)}{T_L} \quad (9)$$

$$a_w = \exp[k_1 - k_2 \exp(-k_3 M)] \quad (10)$$

The second term on the right hand side of Eqs. (4) and (5) (boundary conditions for the heat equation) accounts for the heat of evaporation of the water leaving the surface, where h_{fg} (J/kg) is the water latent heat of evaporation, given by the following equation (ASHRAE, 2009):

$$h_{fg} = 2501.05 \times 10^3 \times \left(\frac{647.3 - T}{647.3 - 273.15} \right)^{0.3298} \quad (11)$$

The concentration of water vapor in air (C_{air}) is derived from the ideal gas law, according to the following equation:

$$C_{air} = 2.1667 \times 10^{-3} \times \frac{RH}{100} \times \frac{p_{vs}(T_{air})}{T_{air}} \quad (12)$$

where p_{vs} (Pa) is the water vapor pressure, given by (ASHRAE, 2009):

$$p_{vs} = \exp \left[\frac{-5.8 \times 10^3}{T} + 1.391 - 4.864 \times 10^{-2} T + 4.176 \times 10^{-5} T^2 - 1.445 \times 10^{-8} T^3 + 6.545 \ln(T) \right] \quad (13)$$

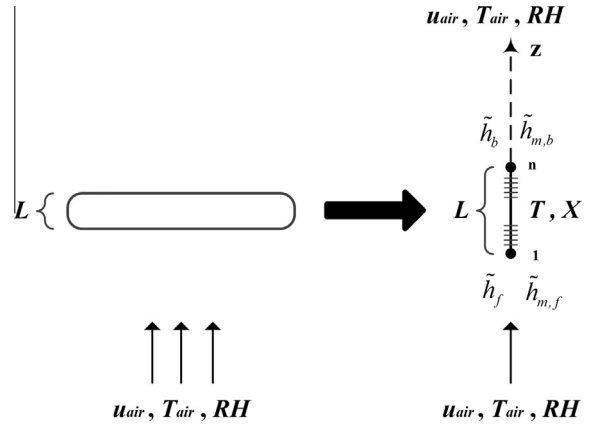


Fig. 1. Flow configuration and one-dimensional numerical model and parameters involved.

The density of the solid material ρ_s , defined by the ratio of the dry material mass to the total volume of the slice, is allowed to vary in the calculations, taking into account the volume shrinkage of the quince slice. A simple mechanistic approach is used to estimate the slice volume variation, based on the assumption that the volume shrinkage at any time step can be approximated to be equal to the removed water volume in that time step (negligible porosity and air mass in the quince slice). The removed water volume is calculated from the moisture change during a time step, which is directly derived by the solution of Eq. (2), while the dry mass is directly derived from the known initial moisture content and the total slice mass, which are both inputs for the numerical model. The approach used for the estimation of volume evolution during drying corresponds to a linear dependence of the product volume on the moisture, which have been recently used in the literature to fit experimental data for potato, with good agreement (Chandra Mohan et al., 2014). The above assumption alone was roughly evaluated for the present case, by comparing the calculated time evolution of the slice volume with measured data.

Eqs. (1) and (2) are numerically discretized using the implicit finite volume method (Patankar, 1980; Ferziger and Peric, 2002) while the derived algebraic equations are solved using the Tridiagonal Matrix Algorithm (Thomas algorithm). Internal iterations are performed in each time step, in order to cope with the non-linearities associated with (i) the coupling of the heat and mass transport equations through the evaporation term in the boundary conditions, (ii) the thermophysical properties dependence on the temperature and moisture and (iii) the dependence of the slice volume and density on the moisture. For the numerical solution of Eqs. (1) and (2), a spatial discretization using 101 grid points (corresponding to a grid size of about 0.06 mm) and a time step equal to 100 s are used, which provide spatial and temporal discretization independency. For the internal iterations within each time step, a convergence criterion equal to 10^{-6} is applied on the calculated average moisture, average temperature and density of the quince slice.

The effective moisture diffusivity D_{eff} is experimentally derived using the well-known slope method, by assuming an Arrhenius type equation (Meziane, 2011; Aghbashlo et al., 2009; Liu et al., 2009). The thermal conductivity and specific heat (Table 2) were estimated as functions of the moisture content (Rao et al., 2005). The heat and mass transfer coefficients were derived from the CFD modeling as described in Section 4.

Table 2

Quince thermophysical properties, as a function of moisture content in wet basis, $M = X/(1 + X)$.

Property	Equation
Thermal conductivity (W/m K)	$k_p(M) = 0.148 + 0.493M$
Specific heat (J/kg K)	$c_p(M) = (1.26 + 2.97M) * 1000$

3. Experimental facility and measurements

The experimental dryer has been previously described in detail by Tzempelikos et al. (2012, 2013). Here, only a brief description is given. The layout of the Laboratory Thermal Convective Dryer (LTCD) is shown in Fig. 2. The drying chamber has dimensions $0.5 \text{ m} \times 0.5 \text{ m} \times 0.66 \text{ m}$. Insulation material (Frelon) of 10 mm thickness has been used to prevent heat losses. The whole system includes a centrifugal fan with backward curved blades, a diffuser with a 6.7° divergence angle to adjust LTCD length with minimum pressure losses, a heat exchanger connected to boiler-hot water system, a spraying system that can feed the air stream with water mist and a set of dampers which allow flow recirculation or new, fresh air to enter into the dryer. A metal tray of 78% permeability is used to align the quince slices in the drying chamber. The design of the whole facility is modular, so that the drying chamber can be vertically or horizontally aligned. For the experiments presented here, the vertical orientation was adopted. Five A class tolerance Pt 100 probes were used for temperature measurements for the product and the drying air. Three humidity transmitters were used for measuring the relative humidity of the drying air. In order to measure the air drying velocity at the inlet of the drying chamber, a pitot tube rake was connected to differential pressure transmitter.

A system of four load cells and analogue transmitter connected to a PC with a DAQ device through different modules provides the option for automated weight loss measurement and data acquisition. The average velocity is adjusted by weighting twelve dynamic pressure measurements using three pitot tube rakes – each one consisting of four pitot tubes – and static pressure measurements at the wall of the duct system, prior to the inlet of the drying chamber. Additional measurements of the weight and the dimensions of samples were taken at selected time steps, out of the dryer. A custom-made virtual instrumentation application was developed in Labview® in order to operate and control the LTCD.

The initial moisture content (M_{in}) was measured by the oven-drying method at 105°C for 24 h (AOAC, 1990) and found to be 81.04% (w.b) or equivalently 4.274 kg water/kg dry (d.b.). The quince slices were placed having its axis parallel to the air-stream. Temperature was measured with a thermocouple Pt100 probe inserted in the quince slice. The LTCD unit was started 2 h before each experiment, in order to achieve steady state conditions. Experiments were performed for drying air temperature of 40, 50 and 60°C and velocity of 1 and 2 m/s. The volumetric flow rate varied from 900 to $1800 \text{ m}^3/\text{h}$, corresponding to a range of Reynolds number from 2.64×10^4 to 5.27×10^4 . The experimental conditions are summarized in Table 3.

The effective moisture diffusivity (D_{eff}) is obtained from the experimental moisture ratio data, using the slope method. Details of the method can be found in (Tzempelikos et al., 2014). A temperature dependence of the effective diffusion coefficient is assumed, following an Arrhenius type equation:

$$D_{eff} = D_0 \exp\left(-\frac{E_a}{R_g T}\right) \quad (14)$$

where D_0 is the pre-exponential factor of the Arrhenius equation (in m^2/s); E_a is the activation energy (in kJ/mol); R_g is the gas constant (in $\text{kJ}/\text{mol K}$); T is the absolute temperature, resulting to the equation:

$$D_{eff} = 5.492 \times 10^{-4} \exp\left(-\frac{4431.713}{T}\right) \quad (15)$$

The correlation of this equation with the experimental data is shown in Fig. 3, while the values of the effective diffusion coefficient for the different experimental cases are given in Table 4.

4. Computational fluid dynamics simulations

Steady-state CFD simulations were performed to estimate the heat and mass transfer coefficients. Flow and temperature fields inside the drying chamber of the LTCD unit were calculated using the commercial CFD code Fluent® (Fluent Inc., 2006). In the initial stage of drying, an unsteady transitional variation of the flow and temperature field around the slice is expected, until steady state conditions are reached. However, this transitional period is very short and there is no practical benefit rather than increased computational cost (Ateeque et al., 2014). For the present geometry, due to the symmetry, the chamber and the cylindrical quince slice

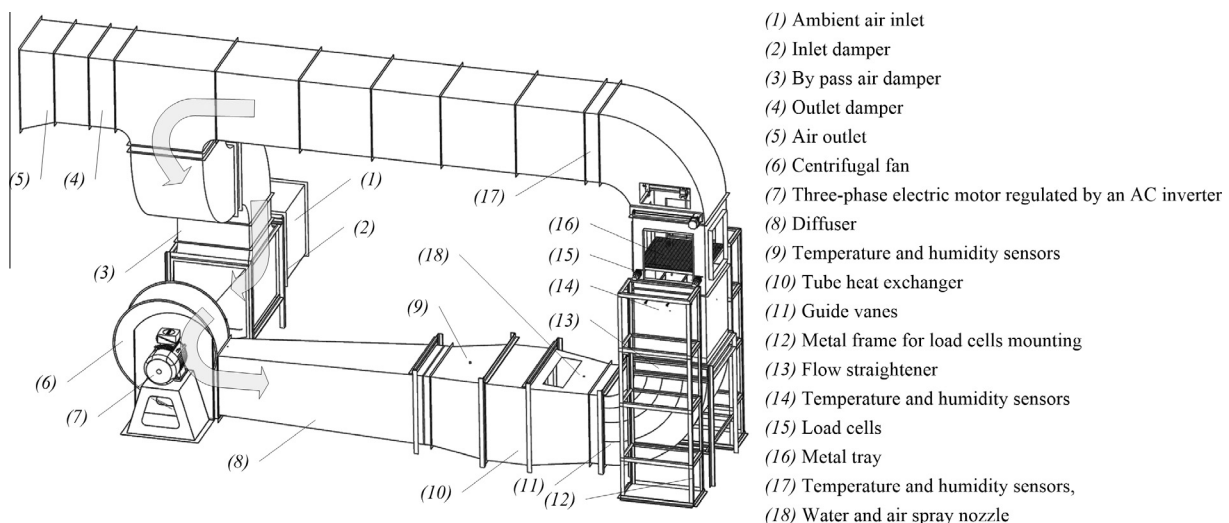
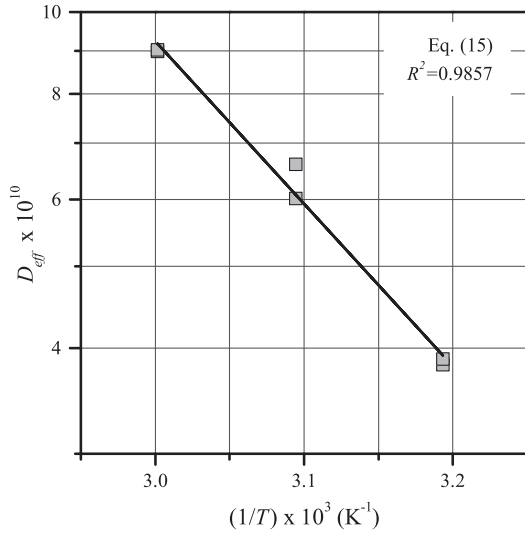


Fig. 2. Schematic diagram of the LTCD unit (curved arrows indicate the flow direction when dryer is in recirculation operation).

Table 3

Conditions for the experimental measurements.

T_{air} (°C)	u_{air} (m/s)	RH (%)	d (m) $\times 10^3$	L_{in} (m) $\times 10^3$	L_{fn} (m) $\times 10^3$	X_{fn} (d.b.)	w_{in} (kg) $\times 10^3$	w_d (kg) $\times 10^3$	T_{in} (°C)
40	1.0	19.54	92.50	9.20	2.50	0.198	61.74	11.71	16.40
50	1.0	12.24	82.80	10.60	2.79	0.583	56.66	10.75	17.70
60	1.0	6.77	89.00	10.58	2.14	0.253	65.43	12.40	20.88
40	2.0	17.14	81.80	9.20	2.06	0.371	48.25	9.15	17.12
50	2.0	9.00	85.00	10.04	2.70	0.150	56.48	10.71	20.65
60	2.0	2.93	87.01	9.50	2.05	0.180	56.28	10.67	24.51

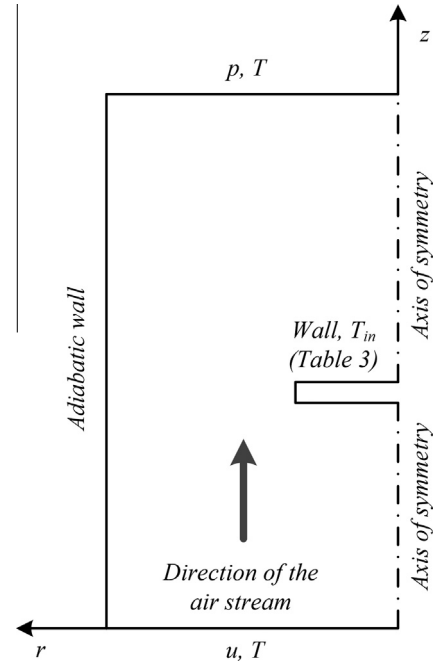
**Fig. 3.** Arrhenius-type relationship between effective moisture diffusivity and temperature.**Table 4**Effective moisture diffusivity coefficient (D_{eff}).

T_{air} (°C)	u_{air} (m/s)	D_{eff} (m ² /s)	R^2
40	1.0	3.82343×10^{-10}	0.9921
50	1.0	6.01866×10^{-10}	0.9901
60	1.0	8.98063×10^{-10}	0.9889
40	2.0	3.88313×10^{-10}	0.9954
50	2.0	6.60007×10^{-10}	0.9894
60	2.0	9.02179×10^{-10}	0.9865

are modeled as a 2-D axis-symmetric configuration, as shown in Fig. 4 taking into account only the half of the chamber and quince. Measured velocity and temperature at the inlet and measured pressure and temperature at the outlet were imposed as boundary conditions (Fig. 4).

The flow around the quince slice is considered as a flow around a bluff body. It is well known that for high Reynolds numbers, turbulence modeling is needed to assess the fluctuations of the flow and thermal fields (Murakami, 1993; Defraeye et al., 2010; Ateeque et al., 2014). In the present work, the SST $k-\omega$ turbulence model is used to account for turbulence (Menter, 1994). This model has been used in the previous works of Defraeye et al. (2010) and Ateeque et al. (2014), in order to reproduce with sufficient accuracy experimental data for external flows.

CFD simulations were performed for all cases listed in Table 3. A structured mesh, with appropriate refinement around the quince slice is used, in order to capture reattachment and wall fluxes towards the quince surface. The mesh consisted of 715 and 218 grid points in the axial and in radial direction, respectively. The wall characteristic length (y^+) was of the order of 1 for all of the

**Fig. 4.** Computational domain.

surfaces of the slice, indicating a sufficient grid resolution in accordance with the turbulence model used.

4.1. Estimation of heat and mass transfer coefficients

The local convection heat transfer coefficient h can be determined from the calculated air temperature field, using the balance between conductive and convective heat transfer:

$$-k_{air} \left(\frac{\partial T}{\partial n} \right)_s = h(T_{ps} - T_{\infty}) \quad (16)$$

where n , is the normal vector at the surface s of the product and T_{∞} is the drying air bulk temperature. After the convective heat transfer coefficient is determined, the convective mass transfer coefficient h_m (m/s) is calculated using the analogy between the thermal and concentration boundary layers (Chilton and Colburn, 1934; Incropera et al., 2012), as commonly made in the literature:

$$h_m = h \left(\frac{D_{va} Le^{1/3}}{k_{air}} \right) \quad (17)$$

where Le , is the Lewis number representing a measure of the relative thermal and concentration boundary layer thickness:

$$Le = \frac{a_{air}}{D_{va}}, \quad (18)$$

a_{air} is the thermal diffusivity of the air and D_{va} is the diffusivity of water vapor in air.

The calculated spatial distribution of the convective heat transfer coefficients exhibits a similar profile in all experimental cases. For this reason, results are presented here for a single experimental case. In Figs. 5 and 6 the spatial distribution of the heat transfer coefficient at the two sides of the slice are shown, for the case of 40 °C and 2 m/s. The values of heat and mass transfer coefficients are in general higher at the front face of the quince. At the front face, the values are larger at the center of the slice and they decrease moving radially towards the edge. At the back face, the heat and mass transfer coefficients present an inverse spatial distribution: the maximum value is located at the edge of the slice, while the minimum near the center.

In Tables 5 and 6, the calculated convective heat and mass transfer coefficients, in terms of their minimum, maximum and average values in both sides of the slice are summarized for all experimental cases. The coefficients were found to be practically independent from the air temperature, but they depict a strong dependency on the air velocity. In particular, the average heat and mass transfer coefficients increase for more than 50%, as the velocity increases from 1 m/s to 2 m/s. This result indicates the relative importance of the air drying velocity on the drying process: higher air velocity result to higher heat and mass transfer coefficients, which in turn induce higher heat and moisture fluxes at the surface and therefore speed up the overall drying process.

The weighted average heat and mass transfer coefficients at the front and back side of the slice, which are necessary for the one-dimensional model of heat and mass transfer inside the quince, are calculated from the local values according to the following equation:

$$\bar{h} = \frac{1}{A} \int h dA \quad (19)$$

$$\bar{h}_m = \frac{1}{A} \int h_m dA \quad (20)$$

The Nusselt number is typically expressed as a function of Pr and Re_p numbers of the flow around the slice:

$$Nu = c Re_p^e Pr^{1/3} \quad (21)$$

In order to obtain more data for the fitting of Eq. (21) and also to extend the range for the application of the proposed correlation, additional CFD simulations were performed for two additional air velocities, namely 0.5 m/s and 1.5 m/s (for 40 °C, 50 °C and 60 °C). The results at both sides of the slice for all the test cases are shown in Fig. 7. The figure includes also the fitting equations. The values of c and e in Eq. (21) for the front and the back face of the cylindrical slice and presented in Table 7.

5. Results and discussions

The approach used in the model for calculating the change of the volume of the slice during drying is roughly evaluated by comparing the calculated slice volume with the measured data. The measured data of the slice volume were obtained by direct measurements of the slice thickness and diameter and assuming a perfect cylindrical geometry. In Fig. 8, the calculated slice volume, as a function of time is shown, in comparison with measured data for the experimental case of 40 °C and 2 m/s. A fine agreement is observed between the calculated and the measured data for the two thirds of the drying time. At the last stages of drying, significant deviations are observed between the model predictions and measurements, which can be attributed to the approach used for measuring the volume which is made geometrically, based on the assumption of a perfect cylinder. This assumption, however,

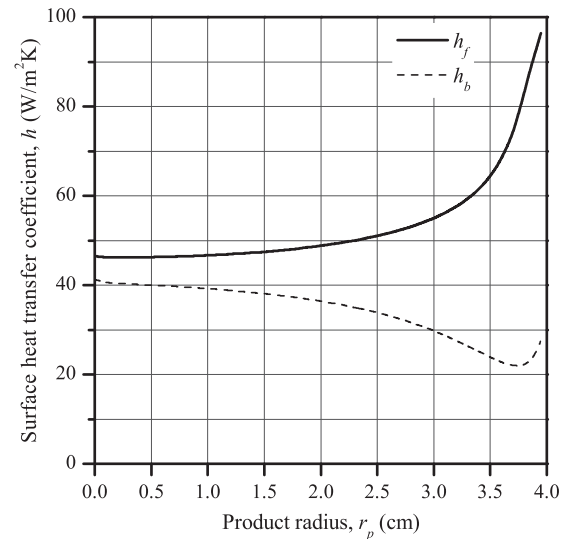


Fig. 5. Calculated heat transfer coefficient at the front and the back face of the quince slice ($T = 40$ °C, $u = 2$ m/s).

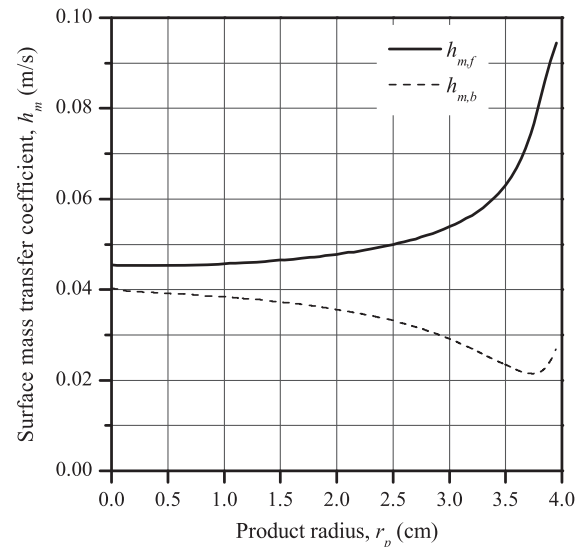


Fig. 6. Calculated mass transfer coefficient at the front and the back face of the quince slice ($T = 40$ °C, $u = 2$ m/s).

Table 5

Maximum, minimum and weighted average heat transfer coefficients at different velocities and temperatures.

Faces	T_{air} (°C)	Heat transfer coefficient, h (W/m ² K)					
		Air velocity, 1.0 m/s			Air velocity, 2.0 m/s		
		Min.	Max.	Average	Min.	Max.	Average
Front	40	27.58	75.96	37.86	46.21	98.53	58.75
Front	50	27.50	76.18	37.91	44.53	99.08	57.01
Front	60	24.29	69.96	34.25	43.81	100.87	56.38
Back	40	14.43	24.12	18.84	21.85	41.18	31.00
Back	50	14.48	24.08	18.86	21.18	39.43	29.66
Back	60	13.20	21.54	17.05	20.94	38.63	29.08

is increasingly inaccurate, in particular in the last period of drying, where the slice is highly deformed.

In order to take further into account in the calculations the shrinkage of the quince, the thickness of the slice was estimated as the average of the initial and the final thickness of the product.

Table 6

Maximum, minimum and weighted average mass transfer coefficients at different velocities and temperatures.

Faces	T_{air} (°C)	Mass transfer coefficient, h_m (m/s)					
		Air velocity, 1.0 m/s			Air velocity, 2.0 m/s		
		Min.	Max.	Average	Min.	Max.	Average
Front	40	0.0270	0.0745	0.0371	0.0453	0.0966	0.0576
Front	50	0.0265	0.0734	0.0383	0.0429	0.0954	0.0576
Front	60	0.0230	0.0662	0.0356	0.0415	0.0955	0.0586
Back	40	0.0141	0.0236	0.0185	0.0214	0.0404	0.0304
Back	50	0.0139	0.0232	0.0190	0.0204	0.0380	0.0299
Back	60	0.0125	0.0204	0.0177	0.0198	0.0366	0.0302

The initial thickness is known, while the final can be approximated *a priori*, based on the assumption that the dimensions of the slice vary only in the axial direction, i.e. only the thickness of the slice changes during the drying process, while the diameter remains constant. The final thickness can be estimated from the ratio of the final slice volume (which is approximated to be the initial volume minus the volume of the initial water content) over the area of the quince slice, which is assumed to remain constant during the drying. Using this approach, the final thickness of the slice is calculated to be about 20% of the initial thickness, a value close to the measured data that range between 20% and 27% of the initial thickness (Table 3).

In Fig. 9, the spatially-averaged moisture content and the temperature at the center of the quince slice are plotted as functions of time, in comparison with the measured data. For the present model, the effective diffusion coefficient given by Eq. (15) was corrected by a factor equal to 0.40, for all the cases examined. The correction of the value of the diffusion coefficient was necessary, due to the different level of approximations inherent in the approach used for the experimental derivation of the diffusion coefficient, in comparison with the present numerical model. It is commonly assumed that the effective diffusion coefficient describes the combined effects of air and liquid transport inside the drying material. So, its value cannot be derived by the mass transport theory (Datta, 2007a, 2007b; Zhang and Datta, 2004); instead, it can only be derived by a semi-empirical method, based on the post-processing of the available experimental data. In the approach used in Section 3, the derivation of the diffusion coefficient is based on the fitting of the analytical solution of the Fick's

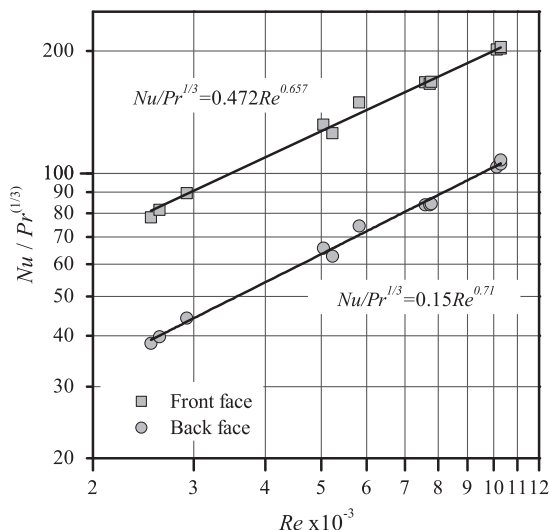


Fig. 7. Correlations for the Nusselt number at opposite faces of cylindrical quince slice.

Table 7

Correlation constants at opposite faces of cylindrical quince slice for air.

Faces	$Nu = c Re_p^e Pr^{1/3}$		R^2
	Constant (c)	Exponent (e)	
Front	0.472	0.657	0.994
Back	0.150	0.710	0.993

diffusion equation to the experimental drying curves (slope method). However, the analytical solution used for the post-experimental processing is a result of a series of approximations (e.g. isothermal conditions, negligible surface resistance, constant properties), attributing 'artificially' the effect of a number of neglected parameters to the diffusion coefficient itself (see also Ruiz-López et al., 2004).

These approximations are not employed in the present numerical model. It is, thus, reasonable that the diffusion coefficient derived by fitting the analytical solution to the measured data needs additional matching, in order to be used in the present more detailed model. In order to achieve higher accuracy between experimental data and predictions a non-linear regression analysis, using an Arrhenius type effective diffusion equation, was also performed. However, preliminary result, obtained using the SQP (Sequential Quadratic Programming) and Interior Point algorithms (available in constrained non-linear optimization toolbox of MATLAB®) for the minimization of the Chi-square function (see also Da Silva et al. (2012)) showed only small improvement of the calculated results (with a significant increase of the computational cost: from less than a minute per run to about 1/2 hour). The limitation on the improvement, even when optimization is used, may be attributed to the formula used for the estimation of the diffusion coefficient. In order to further improve the accuracy of the model, more complicated forms of the diffusion coefficient equation should be employed, by introducing in addition to the temperature, a dependence also on the moisture content (Ateeque et al., 2014; Islam et al., 2003; Datta, 2007b) in the non-linear regression analysis (as also discussed in Ruiz-López et al. (2004)).

From the comparison between the experimental and the numerical results shown in Fig. 9, it can be inferred that the present model is able to predict with acceptable accuracy without the need for any further adjustment or experimental fitting, the effect of the different experimental drying conditions, i.e. the

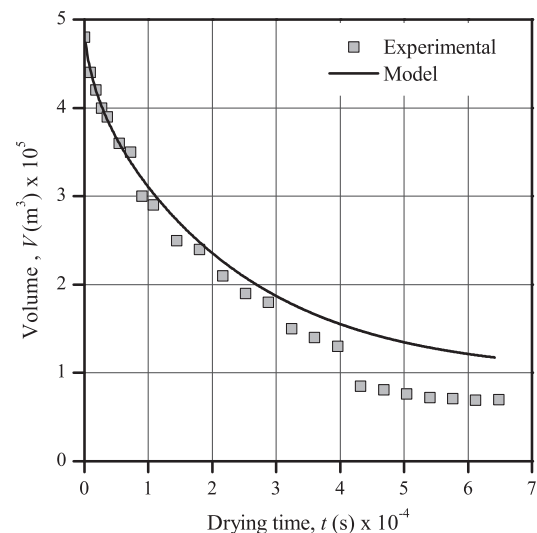


Fig. 8. Time evolution of the volume of the product ($T = 40$ °C, $u = 2$ m/s).

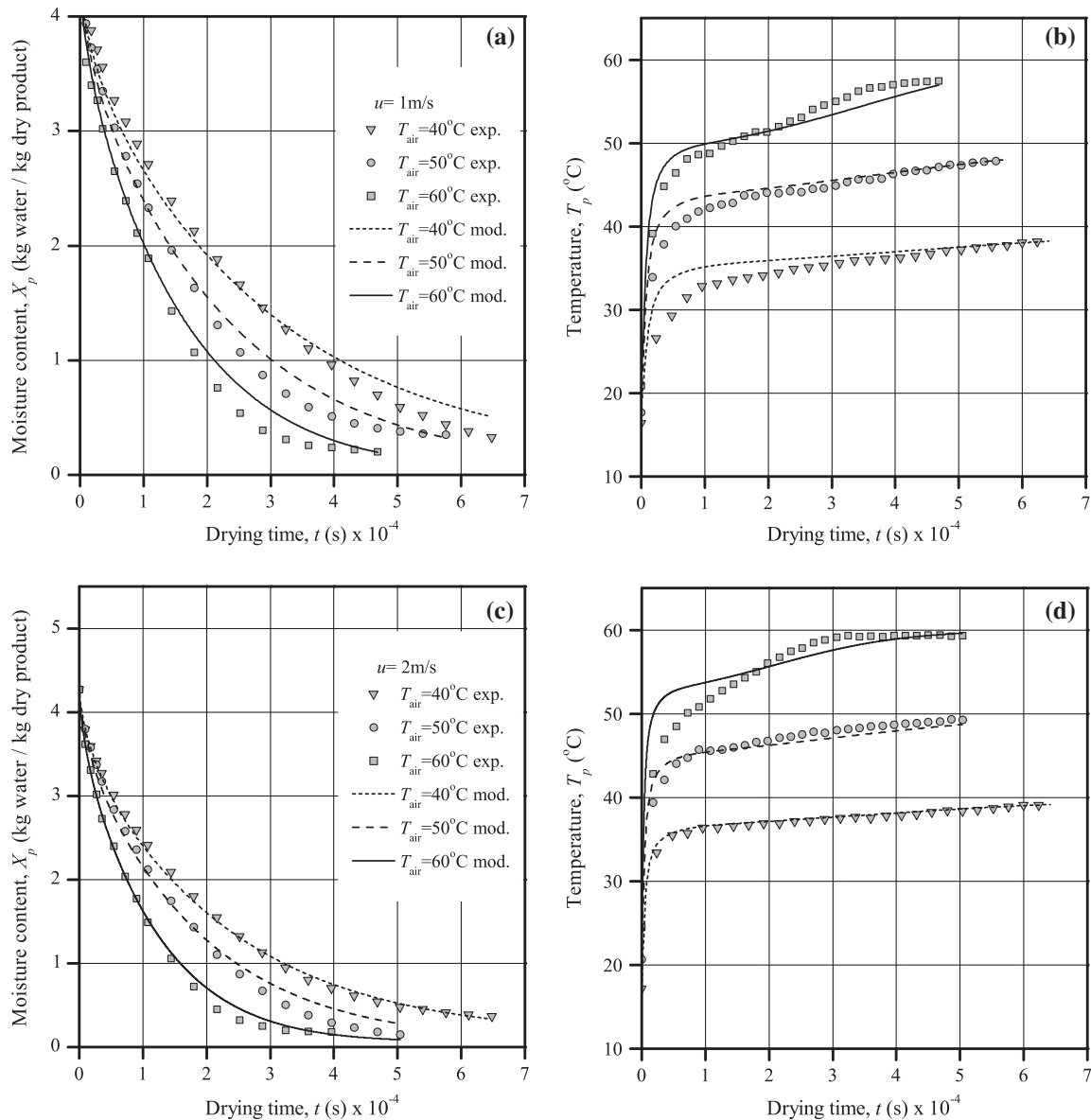


Fig. 9. Calculated moisture (a and c) and temperature (b and d) as a function of time, in comparison with measurements, during the drying process for the conducted experiments.

dependence on the air stream temperature and velocity, provided an appropriate single Arrhenius type diffusion coefficient relation and the appropriate heat transfer coefficient. The deviations observed between the calculated and measured data, can be also attributed to other uncertainties, as for example in relation to the change of the slice geometry and shape during drying that can produce variations of the heat and mass transfer coefficients as the drying process evolves in time, an effect that has not been taken into account in the present CFD simulations.

As mentioned previously, the results in Fig. 9 indicate that water removal is faster for higher drying air temperatures and velocities. This can also be illustrated in Fig. 10, where the time evolution of the drying rate is presented. In this figure, it is shown that up to a critical time where the larger amount of water has been removed, higher drying air temperatures and velocities are associated with higher drying rates. This behavior is due solely to the strong dependence of the moisture effective diffusivity on temperature. For higher temperature differences between air stream and product, the heat flux to the interior of the product is higher

and the effective diffusivity increases. In addition, higher air velocities are associated with higher heat transfer coefficients (Table 5), resulting in turn to faster increase of temperature and thus to higher diffusivity and higher moisture removal rate. As it is discussed later in this paper, the variation of the mass transfer coefficient and the boundary conditions of the moisture equation have negligible influence on the drying process. After the initial period, the pattern inverses and the drying rate is lower for higher air temperatures and velocities, since the higher drying rate at the first period leads faster to such a low moisture content that is more difficult to be removed. This behavior is also confirmed by experiments (Tzempelikos et al., 2015).

The drying rate curves depict the characteristic shape of the so-called falling rate period. The drying rate initially decreases with a high rate within a relatively narrow period of time followed by a more extended period which is characterized by a slower decrease of the drying rate. It is widely accepted that in the falling rate period the drying process is governed by the internal water transport by diffusion (diffusion-controlled), while the external conditions

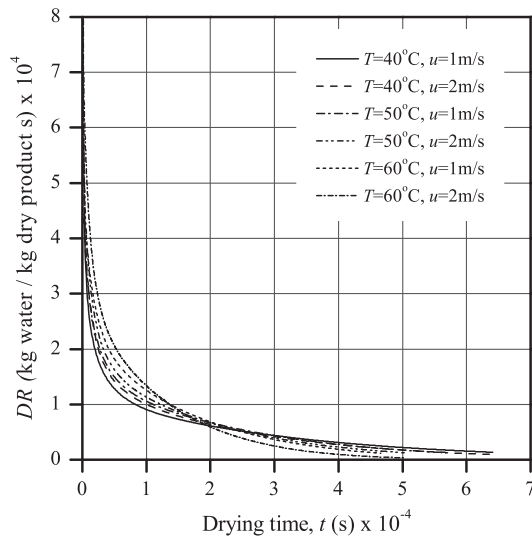


Fig. 10. Calculated drying rate for the different experimental cases.

have practically no influence (Barbosa-Canobas et al., 1996). In the falling rate period, only a very small amount of water exist at the surface of the product because the moisture content at the surface is evaporated very fast, due to the low 'resistance' from the quince material there, while at the same time, the internal moisture migrates (to the surface) due to the concentration gradients. This is also illustrated in Fig. 11a, where the calculated spatial distribution of moisture profile inside the product at various moments during drying is presented. As it is shown, a moisture profile is developed in the interior of the product, while close to the surface boundary the moisture drops almost immediately creating high spatial gradients. This behavior has been also confirmed experimentally for other fruits (Aregawi et al., 2013). The diffusion of the internal water towards the surface, however, is much slower due to the high 'internal mass resistance', resulting to a considerably slower progress of the water removal (Ateeque et al., 2014; Chandra Mohan and Talukdar, 2010). The existence of strong spatial gradients of moisture inside the product is also compatible with the high values of the mass transfer Biot number (of the order of 10^6), which is much higher than the corresponding limit, of the order of 1000 (Parti, 1993). Despite the different values of the mass

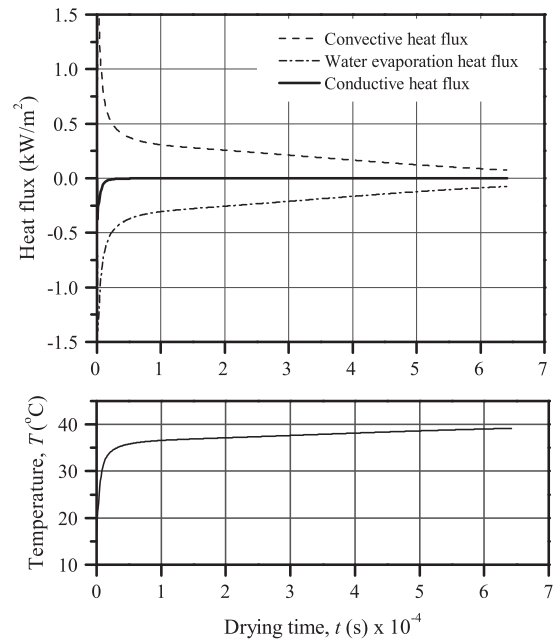


Fig. 12. Calculated heat fluxes and temperature, as a function of time during drying ($T = 40^\circ\text{C}$, $u = 2\text{ m/s}$).

transfer coefficient at the back and the front face of the slice, the moisture profile inside the material exhibits a symmetrical shape, indicating the negligible effect of the external conditions. Similar conclusion have been reached and in the work of (Aregawi et al., 2013) for apples, where a different modeling approach was used.

Fig. 11b shows the spatial distribution of the temperature along the thickness of the quince slice, only for a short period (up to about 1/2 hour) after the beginning of drying, as predicted by the model. The temperature profile presents only small gradients along the quince slice, which vanish quickly. This particular behavior is principally attributed to the low heat transfer Biot number, which range between 0.22 and 0.48. These values are of the same order of magnitude compared to the limit of 'thermally-thin' materials, set to the order of 0.1 (Incropera et al., 2007).

In Fig. 12, the time evolution of the energy balance during drying is depicted, in terms of the calculated heat fluxes, namely the convective heat flux (energy provided from the drying air to the

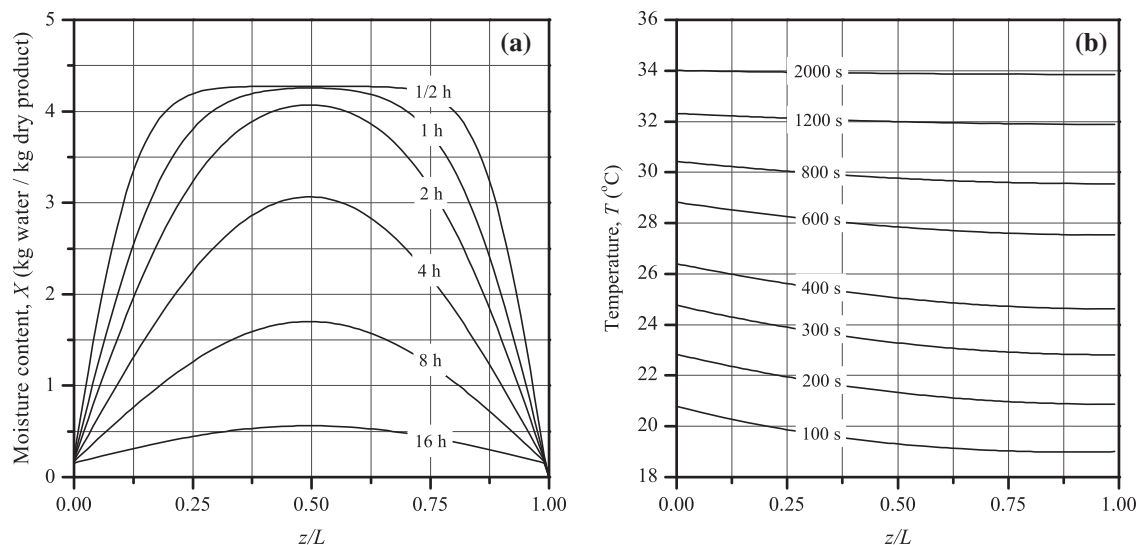


Fig. 11. Moisture (a) and temperature (b) profiles inside the quince slice, for different times during drying ($T = 40^\circ\text{C}$, $u = 2\text{ m/s}$).

quince slice), the water evaporation heat flux (energy consumed for the evaporation of the water at the surface) and the conductive flux (energy conducted in the interior of the quince heating the material). These fluxes correspond to the three terms in the boundary conditions of the heat equation (Eqs. (4) and (5)). The temperature evolution is also shown. As it is shown from this figure, the energy available for the heating of the material is only a very small portion (one order of magnitude lower) of the total energy provided to the product by the air stream. The latter is practically consumed for the water evaporation. At the first short period of drying, however, even this small portion is adequate to heat the material, causing a rapid increase of its temperature. The increase of the temperature results to the decrease of the convective energy provided to the material or equivalently to a decrease of the energy available for further water evaporation and material heating. After this short period and up to the end of the process, almost the entire energy is consumed for water evaporation, resulting to a very slow temperature increase and to the “plateau” depicted in the temperature evolution curve.

6. Conclusions

A 1-D unsteady heat and mass transfer numerical model was developed to study the transient behavior of temperature and moisture content during drying of cylindrical quince slices. The heat transfer inside the quince is considered to be by conduction while the moisture transfer is considered to be governed solely by liquid diffusion. Evaporation is considered to take place only at the surfaces of the quince slice. The averaged heat and mass transfer coefficients required for the model are calculated from CFD simulations. The latter are performed for a steady turbulent flow in a two-dimensional axisymmetric computational domain, incorporating the SST $k-\omega$ turbulence model for the prediction of the external flow and temperature fields around the product. A new correlation is derived for the heat convective transfer coefficient, as function of Reynolds and Prandtl numbers, for the particular flow configuration, namely air flow parallel to the axis of cylindrical slices, which is scarcely considered in the literature. The effective diffusivity is estimated experimentally using a typical method that involves a simplified analytical solution of diffusion inside the product.

In diffusion-based model the various complex moisture transport mechanisms in the product, which still are not fully resolved in drying theory and modeling, are surrogated by a simple Fick's diffusion transport model. Due to the approximations of the slope method that is used for the experimental derivation of the effective diffusivity, its value in the model was corrected by a factor of 0.4. Comparisons with experimentally acquired data were performed for a range of flow and thermal conditions demonstrating that the present, diffusion-based model is able to describe the strong heat and mass transfer coupling and capture the time evolution of moisture content and temperature with minimum experimental adjustment. The model is robust, computationally efficient (trivial for a modern PC), and so it is considered suitable for analyzing the drying process, from an engineering point of view.

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